AJAY CHOUDHARY SIR • CHEMISTRY ACADEMY
8+ Years Teaching Excellence • Organic | Inorganic | Practical Chemistry

Unit Test 01: Hydrocarbons & Reaction Mechanisms

Class 11 & 12 Chemistry Assessment (Max Marks: 40, Time: 90 Mins)

Chapter / Focus Area: Organic Chemistry - Hydrocarbons & Alkyl Halides

Section A: Multiple Choice Questions (1 Mark Each)

Q1. Which of the following carbocations is most stable?

(a) $(CH_3)_3C^{(+)}$ (b) $(CH_3)_2CH^{(+)}$ (c) $CH_3CH_2^{(+)}$ (d) $CH_3^{(+)}$

Q2. The ozonolysis of 2-butene followed by Zn/H_2O gives:

(a) Methanal (b) Ethanal (c) Propanal (d) Butanone

Section B: Short Answer Questions (2-3 Marks Each)

Q3. State and explain Markovnikov's rule with a suitable reaction mechanism of propene with HBr.

Q4. How will you convert benzene to nitrobenzene and then to aniline? Give complete reagents and conditions.

Q5. Explain hyperconjugation in propene and draw all possible no-bond resonance structures.

Section C: Long Answer Problem (5 Marks)

Q6. An organic compound (A) with molecular formula C_4H_9Br on reaction with alcoholic KOH forms compound (B). Compound (B) on ozonolysis gives two moles of formaldehyde and one mole of acetone. Identify (A) and (B) and write all steps involved.